22. Using the method of least squares, find the cubic polynomial of the form $y = ax^3 + b$ which best fits the three data points $(-1, 1)$, $(0, -7)$ and $(1, 7)$.

Give the answer in the form $y = ax^3 + b$ with the coefficients a, b in simplest form.

$y = 3x^3 + \frac{1}{3}$

Want to minimize the function (sum of squares of errors)

$$S(a,b) := \underbrace{\left(1 - (a(-1)^3 + b)\right)^2}_{\text{sq. of error at } x=-1} + \underbrace{\left(-7 - (a(0)^3 + b)\right)^2}_{\text{sq. of error at } x=0} + \underbrace{\left(7 - (a(1)^3 + b)\right)^2}_{\text{sq. of error at } x=1}$$

(1 = y-val, -1 = x-val)

$$= (1 + a - b)^2 + (-7 - b)^2 + (7 - a - b)^2$$

$$\frac{\partial S}{\partial a}(a,b) = 2(1 + a - b)(+1) + 2(7 - a - b)(-1) = -12 + 4a$$

$$\frac{\partial S}{\partial b}(a,b) = 2(1 + a - b)(-1) + 2(-7 - b)(-1) + 2(7 - a - b)(-1) = -2 + 6b$$

At min point, $\frac{\partial S}{\partial a}(a,b) = 0$, i.e. $-12 + 4a = 0$, i.e. $a = 3$

$\frac{\partial S}{\partial b}(a,b) = 0$ $\quad -2 + 6b = 0$ $\quad b = \frac{1}{3}$

∴ eqn is $y = 3x^3 + \frac{1}{3}$

23. Use Lagrange multipliers to find the maximum value of the function $f(x,y) = 2x + y$ subject to the constraint $2x^2 + y^2 = 9$.

Give the answer in the form $a\sqrt{b}$ where both a and b are positive integers, and b is as small as possible.

$3\sqrt{3}$

Want max value of $f(x,y) = 2x + y$ under constraint $\underbrace{2x^2 + y^2}_{g(x,y)} = 9$ (*)

Lagrange: at (x,y) giving max value,

there exists $\lambda \in \mathbb{R}$ s.t. $(\nabla f)(x,y) = \lambda \cdot (\nabla g)(x,y)$ (vector eqn)

i.e. $\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \lambda \cdot \begin{pmatrix} 4x \\ 2y \end{pmatrix}$

so $\left.\begin{aligned} 4\lambda x &= 2 \\ 2\lambda y &= 1 \end{aligned}\right\}$ so $x = y$.

Then (*) gives $3x^2 = 9$,

so $x = \pm\sqrt{3}$,

$y = \pm\sqrt{3}$ corresp.

$\rightsquigarrow (x,y) = +(\sqrt{3}, \sqrt{3})$ or $-(\sqrt{3}, \sqrt{3})$

→ for max value of $f(x,y) = 2x + y$ must take $(x,y) = +(\sqrt{3}, \sqrt{3})$

f-value $= 3\sqrt{3}$

24. Find the point (x, y, z) which maximizes the function $f(x, y, z) = xyz$ subject to the constraints

$x > 0,\ y > 0,\ z > 0$ and $x^2 + y^2 + z^2 + 2xz = 18.$

$(\sqrt{3}, \sqrt{6}, \sqrt{3})$

Want to maximize $f(x,y,z) = xyz$ under constraint $x > 0,\ y > 0,\ z > 0$

$$\underbrace{x^2 + y^2 + z^2 + 2xz}_{g(x,y,z)} = 18 \quad \text{—— (4)}$$

Lagrange: at (x,y,z) giving max value of $f(x,y,z)$,

there exists $\lambda \in \mathbb{R}$ s.t. $(\nabla f)(x,y,z) = \lambda \cdot (\nabla g)(x,y,z)$

i.e. $\begin{pmatrix} yz \\ xz \\ xy \end{pmatrix} = \lambda \cdot \begin{pmatrix} 2x+2z \\ 2y \\ 2z+2x \end{pmatrix}$

so

$$\begin{aligned} yz &= \lambda \cdot (2x+2z) && \text{—— (1)} \\ xz &= \lambda \cdot 2y && \text{—— (2)} \\ xy &= \lambda \cdot (2z+2x) && \text{—— (3)} \end{aligned}$$

① & ③ gives $yz = xy$, so $x = z$

Then ② gives $x^2 = 2dy$ ——— ④

① gives $yx = 4dx$, so $y = 4\lambda$

so ④ gives $x^2 = \frac{y^2}{2}$, so $y = \sqrt{2}\,x$

Hence ✳ gives $x^2 + 2x^2 + x^2 + 2x^2 = 18$

$x^2 + y^2 + z^2 + 2xz = 18$

so $x^2 = 3$, so $x = \sqrt{3}$

$y = \sqrt{6}$

$z = \sqrt{3}$

so max occurs at $(\sqrt{3}, \sqrt{6}, \sqrt{3})$

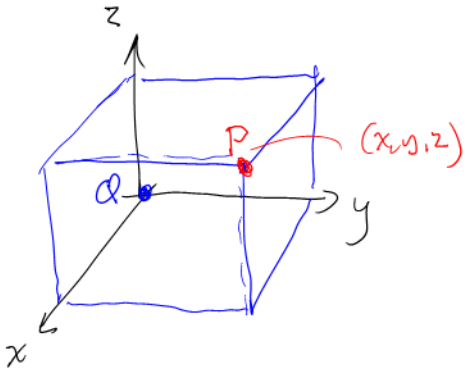
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$x>0,\ y>0,\ z>0$

A rectangular box is placed in the first octant so that one corner Q is at the origin and the three sides adjacent to Q lie in the coordinate planes. The corner P diagonally opposite Q lies on

a) the plane $x + 2y + 3z = 18$ b) the ellipsoid $x^2 + 2y^2 + 4z^2 = 12$.

Using Lagrange multipliers, tell for which point P the box will have the largest volume, (and tell how you know it gives a maximum point.)



a) Want to maximize $f(x,y,z) = xyz$ (volume)

under constraint $\underbrace{x + 2y + 3z = 18}_{g(x,y,z)}$ $(x>0,\ y>0,\ z>0)$

Lagrange multipliers: at max for f, have $\lambda \in \mathbb{R}$ s.t.

$$\nabla f = \lambda \cdot \nabla g$$ (& $x + 2y + 3z = 18$ & $x>0,\ y>0,\ z>0$)

i.e. $$\begin{pmatrix} yz \\ xz \\ xy \end{pmatrix} = \lambda \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

i.e. $\lambda = yz = \frac{xz}{2} = \frac{xy}{3}$

i.e. $y = \frac{x}{2}$, $\frac{z}{2} = \frac{y}{3}$, $z = \frac{x}{3}$

$x = 2y$ $\qquad$ $x = 3z$

so $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3z \\ 3z/2 \\ z \end{pmatrix} = z \begin{pmatrix} 3 \\ 3/2 \\ 1 \end{pmatrix}$

Then $\boxed{x + 2y + 3z = 18}$ gives $3z + 3z + 3z = 18$, so $z = 2$, $y = 3$, $x = 6$

i.e. $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \\ 2 \end{pmatrix}$ is coord for P

b) Want to maximize $f(x,y,z) = xyz$ (volume)

under constraint $\underbrace{x^2 + 2y^2 + 4z^2 = 12}_{g(x,y,z)}$ $\quad (x>0,\ y>0,\ z>0)$

Lagrange multipliers : at max for f, have $\lambda \in \mathbb{R}$ s.t.

$\boxed{\nabla f = \lambda \cdot \nabla g}$ $\quad$ (& $\boxed{x^2 + 2y^2 + 4z^2 = 12}$ & $x>0,\ y>0,\ z>0$)

i.e. $\begin{pmatrix} yz \\ xz \\ xy \end{pmatrix} = \lambda \cdot \begin{pmatrix} 2x \\ 4y \\ 8z \end{pmatrix}$

i.e. $yz = \lambda 2x \longrightarrow \times x$
$xz = \lambda 4y \longrightarrow \times y$
$xy = \lambda 8z \longrightarrow \times z$

get $xyz = 2\lambda x^2 = 4\lambda y^2 = 8\lambda z^2$ — (*)

so $x^2 = \frac{xyz}{2\lambda}$, $2y^2 = \frac{xyz}{2\lambda}$, $4z^2 = \frac{xyz}{2\lambda}$

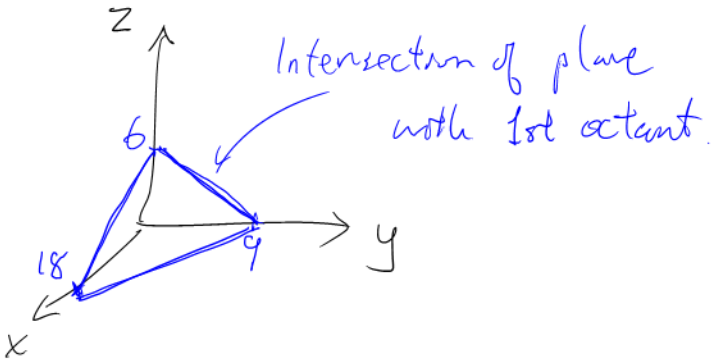
so $\boxed{x^2 + 2y^2 + 4z^2 = 12}$ gives $3 \cdot \frac{xyz}{2\lambda} = 12$, so $\underline{xyz = 8\lambda}$

so (*) gives $8 = 2x^2 = 4y^2 = 8z^2$

so $x>0$, $y>0$, $z>0$ gives $x = \sqrt{4} = 2$
$y = \sqrt{2}$
$z = \sqrt{1} = 1$

i.e. $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ \sqrt{2} \\ 1 \end{pmatrix}$ is where P is.

how you know it gives a maximum point.)



eg Case (a) maximize $f(x,y,z) = xyz$

under Constraint $\boxed{x + 2y + 3z = 18}$

Relax condition slightly to $x \geq 0,\ y \geq 0,\ z \geq 0$

Then on $S := \{(x,y,z) \in \mathbb{R}^3 : x \geq 0,\ y \geq 0,\ z \geq 0 \text{ & } \boxed{x + 2y + 3z = 18}\}$

is closed and bounded

Thm in analysis (eg. MA2108): there exists $(x,y,z) \in S$ such that $f|_S$ attains its max value there.

i.e. for all $(x',y',z') \in S$, get $f(x',y',z') \leq f(x,y,z)$

In our case, Lagrange found for us $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 2 \end{pmatrix}$ either max or min $f|_S$

provided that $x, y, z > 0$

Claim: This is max. for $f|_S$

Pf If not max, then there exists $(x_0, y_0, z_0) \in S$ which maximizes $f|_S$

If this (x_0, y_0, z_0) satisfies $x, y, z > 0$, then Lagrange would have found it for us!

So if $(2, 3, 6)$ found by Lagrange is not max,

then (x_0, y_0, z_0) must violate $x_0, y_0, z_0 > 0$

ie, must have $x_0 = 0$ or $y_0 = 0$ or $z_0 = 0$

ie. (x_0, y_0, z_0) lies on coord planes

But at coord planes, f takes value 0, $\leq$ our max value $2 \cdot 3 \cdot 6 = 36$

contradiction.

Hence $(2, 3, 6)$ found by Lagrange is max, #